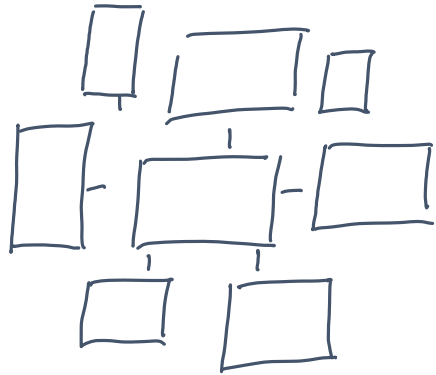




1. Embedded Keyword Spotting on Arm Cortex-M55 with CNN Acceleration
2. Multi-Channel Audio Expansion Board for the STM32G474RE Platform
3. Development of a TinyUSB-Based USB Device Framework on STM32G474RE

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Embedded Keyword Spotting on Arm Cortex-M55 with CNN Acceleration

“Hey Siri”, “Alexa”



Keywords: Edge AI, Machine Learning, Embedded Software, DSP

This project investigates the implementation of **keyword spotting** on modern microcontrollers based on the **Arm Cortex-M55**, including the use of dedicated **CNN accelerators**. The goal is to design and evaluate an end-to-end audio processing pipeline for real-time, always-on speech recognition under strict constraints on latency, power consumption, and memory usage. The project covers the complete signal-processing and machine-learning chain, from raw audio acquisition to classification, and compares software-only, ISA-optimized, and accelerator-based implementations. A particular focus is placed on the Helium/MVE vector extension of the Cortex-M55, which provides ISA-level acceleration for DSP and ML workloads.

Objectives: The main objective is to develop and analyze a complete embedded KWS system, including:

- Real-time audio acquisition and streaming processing
- Feature extraction using STFT, Mel filterbanks, log-Mel spectrograms, and MFCCs
- Deployment of a compact CNN-based KWS model
- Optimization for low latency, low power, and robust operation
- Comparison of software-only, ISA-optimized, and hardware-accelerated approaches

System Architecture: The system follows a streaming pipeline. Audio data is continuously sampled, buffered via DMA, and segmented into short overlapping frames. An Short-time Fourier Transform (STFT) generates a time-frequency representation, which is then mapped onto a Mel filterbank and logarithmically compressed to obtain a log-Mel spectrogram. This representation can be interpreted as a compact 2D feature tensor for CNN input; alternatively, MFCCs may be used as a more compressed feature set.

The resulting features are processed by a compact CNN for keyword classification. To reduce computational load, an additional lightweight front-end stage, such as energy-based detection or voice activity detection, may be used to activate the CNN only when relevant acoustic activity is present.

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Design Constraints: A key focus of the project is optimization under realistic embedded constraints:

- Low latency: fast response through frame-based streaming and efficient pipeline execution
- Low power: always-on operation with DMA-based acquisition, quantized models, and selective activation of compute-intensive stages
- Robustness: reliable operation in the presence of noise, reverberation, speaker variability, and changing signal levels

Implementation Approach: A first implementation will realize the complete pipeline on the Cortex-M55 using optimized DSP and neural-network kernels. This software-based approach will make explicit use of Helium/MVE, SIMD processing, cache-aware programming, and efficient streaming data handling. In a second step, the CNN inference will be offloaded to a dedicated accelerator, while the CPU remains responsible for audio acquisition, feature extraction, and system control. Both approaches will then be compared in terms of performance, latency, and energy efficiency.

Possible Target Platforms: Possible target platforms include modern AI-capable microcontrollers such as the STM32N6 or the Infineon PSOC Edge E84, both of which are suitable for evaluating Cortex-M55-based KWS with and without hardware acceleration.

Multi-Channel Audio Expansion Board for the STM32G474RE Platform

Hardware PCB Design

Keywords: PCB Design with KiCad

This project focuses on the design and development of a multi-layer **audio add-on board** for the **STM32 NUCLEO-G474RE** using **KiCad**. The board will integrate an audio codec, a multi-channel **MEMS microphone array**, and a complete analog input and output signal path. The aim is to create a flexible hardware platform for embedded audio processing, including beamforming, audio analysis, and machine-learning-based acoustic applications.

Objectives: The main objective is to design a compact and high-quality audio expansion board that extends the capabilities of the STM32G474RE. The project includes:

- Design of a multi-layer PCB in KiCad
- Integration of an audio codec for audio input and output
- Integration of 9 or more MEMS microphones for multi-channel acquisition, for example digital MEMS microphones based on PDM, I2S, or TDM interfaces
- Development of a complete analog line-in and line-out path
- Support for DSP, beamforming, and machine-learning-based audio applications

Proposed System Design: The board will include a MEMS microphone array with 9 or more microphones arranged in a defined geometry to support beamforming and direction-of-arrival estimation. Another audio codec will provide high-quality digital audio input and output for recording and playback. Moreover, the board will also include a complete analog signal chain with a 3.5 mm line input, input protection, high-pass filtering, level adjustment, anti-aliasing filtering, for example using a Sallen-Key stage, and proper buffering for the STM32 ADC. A corresponding analog output stage will also be implemented for DAC-based playback, including filtering and line-level output.

Proposed Use Cases: The platform can be used for multi-channel audio acquisition, beamforming, and sound source localization. It also enables a wide range of experiments in embedded DSP and audio system design.

In addition, the board can serve as a hardware platform for machine-learning applications such as acoustic event detection, speech activity detection, and sound classification based on microphone data.

Development of a TinyUSB-Based USB Device Framework on STM32G474RE

Embedded Software Design

Keywords: STM32, TinyUSB

This project is about developing a flexible USB device interface for the **STM32 NUCLEO-G474RE** using **TinyUSB**. The aim is to establish reliable communication between the embedded system and a host PC while supporting different USB device classes. TinyUSB is a lightweight open-source USB stack for microcontrollers. It offers ready-to-use support for standard USB classes such as CDC, HID, and Audio, making USB integration significantly easier than implementing low-level USB drivers from scratch.

Objectives: The main objective of this project is to integrate TinyUSB on the STM32G474RE and use it to implement several practical USB device classes. In addition, the project aims to enable real-time data exchange between the microcontroller and a host PC and to create a reusable USB software framework that can serve as a basis for future embedded developments.

Proposed Use Cases:

A first use case is the implementation of a USB CDC interface, which allows the board to appear as a virtual COM port on the host PC. This can be used for debugging, streaming DSP data, or providing a simple command-line interface for configuration and testing.

Another interesting use case is a USB Audio device, which would allow the system to stream microphone signals or processed audio directly to the PC. This creates a strong link to audio and DSP applications, such as filtering, beamforming, or real-time audio analysis.

The project can also include a USB HID interface for simple and driverless communication with the host. Such an interface could be used to send control signals, trigger commands, or transmit event-based information generated by buttons, sensors, or audio detection algorithms.

Finally, the project may explore a composite USB device, combining multiple USB classes in a single implementation. For example, a CDC interface could be used for debugging and control, while an Audio interface is used for real-time streaming.